Follow the chain from force to position

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Force and motion / ES: fuerza y movimiento / PT: força e movimento

Net force → a → velocity and position.

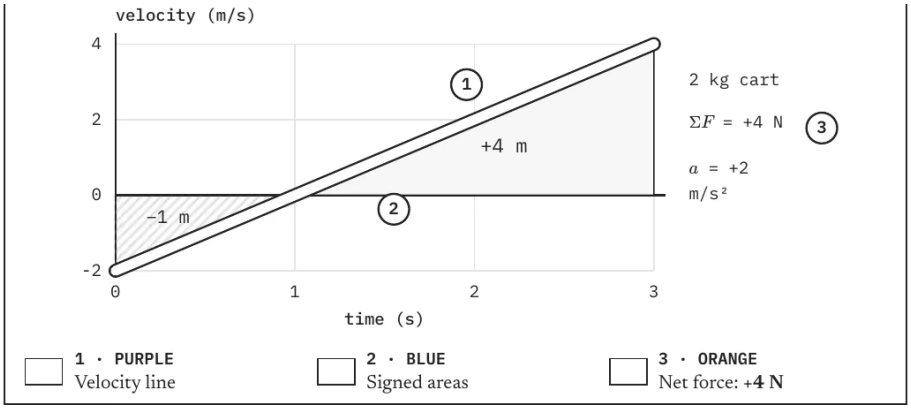
Also on your note page: Interval, Reversal, Area.

B READ THE RELATIONSHIP

Force → acceleration → velocity → position

As you read, find: force · velocity · position.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg cart starts at 1 m with velocity −2 m/s . A constant +4 N net force acts for 3 s . Find final velocity and position.

Given m = 2 kg ; Σ F = +4 N x 0 = 1 m ; v i = −2 m/s t = 3 s

Rule Find a from force. Then steady-a equations.

Work a = 4/2 = +2 m/s² v f = −2 + 2(3) x f = 1 − 6 + 9

Answer +4 m/s ; +4 m Δ x = +3 m ; distance = 5 m .

E SAVE FOR THE FINAL MEETING

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

Read the physics: Force and motion

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Divide net force by mass to find acceleration. For constant acceleration, use the initial velocity and elapsed time to find final velocity.

02 HOW IT WORKS

Use the signed area under the velocity–time graph for displacement. Add displacement to starting position. A section below zero subtracts from displacement but still counts toward distance.

03 WHAT TO CHECK

With zero net force a moving object continues at its initial velocity. With constant nonzero net force its velocity changes by the same amount each second.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg cart starts at 1 m with velocity -2 m/s . A constant $+4$ N net force acts for 3 s . Find final velocity and position. $+4$ m/s ; $+4$ m $\Delta x = +3$ m ; distance $= 5$ m .

Model check: Force → acceleration → velocity → position.

05 CONNECT TO XJ

XJ's forecast needs its present motion and the forces changing it. Use one reference frame through the entire calculation.

SAY IT BEFORE YOU WRITE

What do you need besides force and mass to predict position? From force I can find ___. To find position I also need ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Force and motion

ES: fuerza y movimiento / PT: força e movimento

Meaning in my words	How can you tell?
How does a net force change motion? In my words, this means ___.	A cart feels a net force right. What else tells you whether it is speeding up right now? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch a cart moving left while a net force points right. Which way does acceleration point? My example shows ___.	"The force points right, so the cart is already going right." What evidence is missing? I would change ___ because ___.

Interval

ES: intervalo de tiempo / PT: intervalo de tempo

Meaning in my words	How can you tell?
How would you explain interval to a friend? In my words, this means ___.	The clock reads 3 s at the start and 7 s at the end. How long is the interval? I can tell because ___.
Example or sketch	**Non-example and why**
You are using velocities from 3 s and 7 s. Which strip of the graph belongs in your calculation? My example shows ___.	A partner uses 7 s as the elapsed time. Which part did they include by accident? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Reversal

ES: inversión del sentido / PT: inversão do sentido

Meaning in my words	How can you tell?
How would you explain reversal to a friend? In my words, this means ___.	On a velocity–time graph, how can you tell a cart actually reversed? I can tell because ___.
Example or sketch	**Non-example and why**
A cart goes from +2 m/s through zero to −1 m/s. Draw its direction before and after. My example shows ___.	A bus stops, then keeps going east. Does stopping prove it reversed? I would change ___ because ___.

Area

ES: área / PT: área

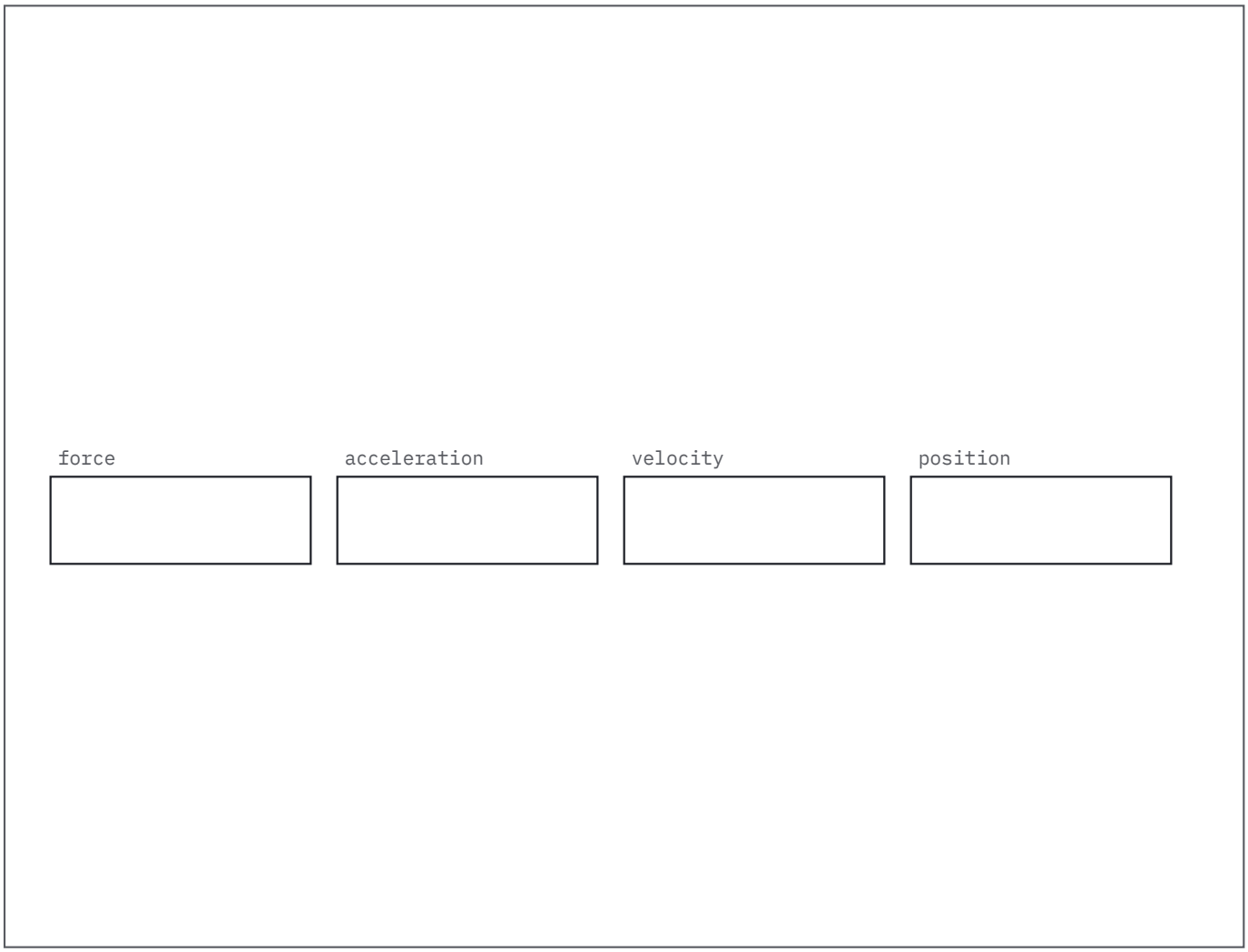
Meaning in my words	How can you tell?
How would you explain area to a friend? In my words, this means ___.	What do you do with the area below the time axis when you find displacement? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show? My example shows ___.	Someone labels the area −2 m/s. Which graph reading did they use? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Build a four-frame chain from force to position. Add the missing inputs and units beside the arrows.

Labels you can use: force · velocity · position.



Tell a partner

From force I can find ___. To find position I also need ___.

Can we release a forecast?

20

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Identify which measurements a position forecast needs.

CHOOSE ONE JOB TOGETHER

MISSION SYSTEMS ENGINEER ☐

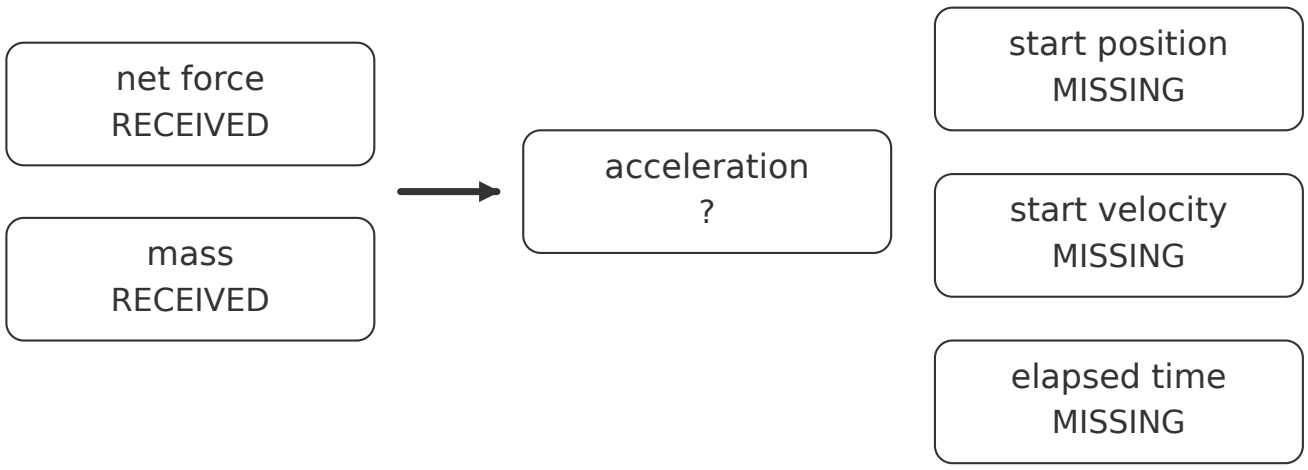
Mark what you can calculate now. Send the telescope operator a request for the measurements still needed.

NEWS ASSIGNMENT EDITOR ☐

Prepare a 'known / still needed' graphic. Tell the anchor which result can air and which must wait.

Word help: known / missing / forecast

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.



01 OUR FIRST CALL

What can the desk release now? What measurements must the crew still collect?

We can find ___. We still need ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Is the forecast ready after the update? Say what you can release and what is still missing.

Now we would say ___ because ___.

Switch speakers. If XJ reverses direction during the forecast, how would that affect a report of distance versus displacement?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks